
SOFTWARE Manual

for

VMM3 Slow Control

Version 1.0

Prepared by Michael Lupberger, Manuel Guth

CERN, RD51

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Contents

1. Introduction	3
1.1. About RD51	3
1.2. About VMM3(a)	3
2. Getting Started	5
2.1. Requirements	5
2.2. Installation	5
2.2.1. Installing QT	5
2.2.2. Obtaining the software	5
2.2.3. Compiling the software	6
3. Software Registers	7
3.1. VMM Slow Control	8
3.1.1. DAQ Level	9
3.1.2. FEC Level	10
3.1.3. HDMI Level	11
3.1.4. Hybrid Level	12
3.1.5. VMM Level	13
3.1.6. Config Module	15
3.2. Calibration	15
3.3. Logging Window	16
4. Examples	17
4.1. Test Pulse Setup	17
A. VMM Resgister details	19

1. Introduction

This software, based on QT, provides a GUI for the VMM slow control. The VMM [1] is a new readout chip developed for the New Small Wheel [2] update of the ATLAS [3] experiment at CERN.

The basic framework is the VMM read out software developed by ATLAS which can be found in the CERN gitlab repository https://gitlab.cern.ch/NSWelectronics/vmm_readout_software (only internally accessible). The user interface is completely revised for more flexible purposes also outside of the ATLAS NSW project as well as the data handling in the background.

1.1. About RD51

«The proposed R&D collaboration, RD51, aims at facilitating the development of advanced gas-avalanche detector technologies and associated electronic-readout systems, for applications in basic and applied research. The main objective of the R&D programme is to advance technological development and application of Micropattern Gas Detectors. The invention of Micro-Pattern Gas Detectors (MPGD), in particular the Gas Electron Multiplier (GEM), the Micro-Mesh Gaseous Structure (Micromegas), and more recently other micro pattern detector schemes, offers the potential to develop new gaseous detectors with unprecedented spatial resolution, high rate capability, large sensitive area, operational stability and radiation hardness. In some applications, requiring very large-area coverage with moderate spatial resolutions, more coarse Macro-patterned detectors, e.g. Thick-GEMs (THGEM) or patterned resistive-plate devices could offer an interesting and economic solution. The design of the new micro-pattern devices appears suitable for industrial production. In addition, the availability of highly integrated amplification and readout electronics allows for the design of gas-detector systems with channel densities comparable to that of modern silicon detectors. Modern wafer post-processing allows for the integration of gas-amplification structures directly on top of a pixelized readout chip. Thanks to these recent developments, particle detection through the ionization of gas has large fields of application in future particle, nuclear and astro-particle physics experiments with and without accelerators.

The RD51 collaboration involves around 450 authors, 75 Universities and Research Laboratories from 25 countries in Europe, America, Asia and Africa. All partners are already actively pursuing either basic- or application-oriented RD involving a variety of MPGD concepts. The collaboration established common goals, like experimental and simulation tools, characterization concepts and methods, common infrastructures at test beams and irradiation facilities, and methods and infrastructures for MPGD production.» [4]

1.2. About VMM3(a)

The VMM is a new custom Application Specific Integrated Circuit (ASIC) developed by the ATLAS collaboration at CERN for the new front end readout electronics of the New Small Wheel (NSW) upgrade. The development was done at Brookhaven National Laboratory by Gianluigi de Geronimo and his microelectronics design group.

The VMM3 is the third generation of prototypes and the second last step before the mass production. After resolving some issues of the current version the VMM3a will be the final

design. The VMM architecture is shown in figure 1.

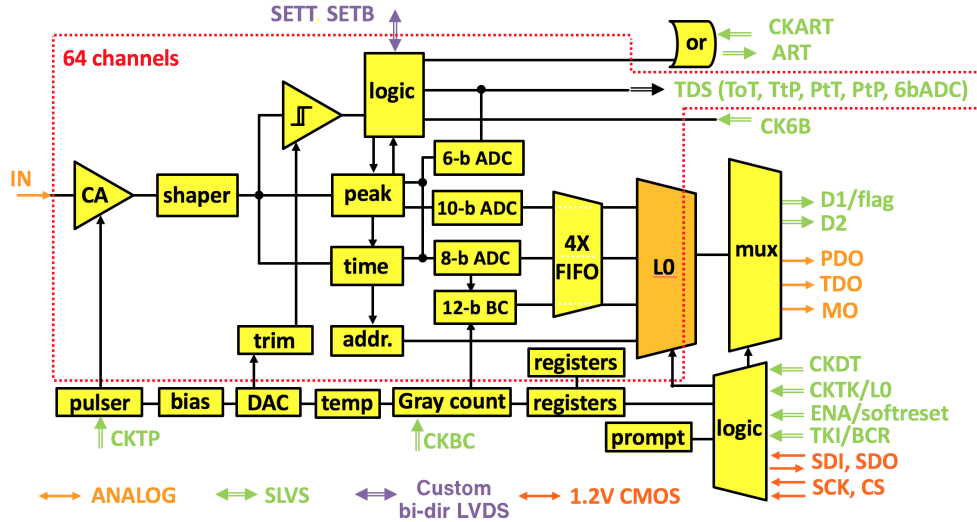


Figure 1: VMM3 architecture [5]

The RD51 collaboration developed a Scalable Readout System (SRS) [6] which allows the user to choose the frontend (ASIC, hybrid) that fits best for the detector. It is a point-point like technology. Furthermore, it is possible to scale minimal start-up systems to a large system. The typical combination consists of a FEC and an adapter card where the adapter card depends on the frontend. The development of the hybrid and the adapter card customised for the VMM is also done within the RD51 collaboration.

Each hybrid holds two VMM chips. The full schematic of the detector electronics chain is shown in figure 2. The software is designed as a kind of projection of this detector electronics chain from the VMM over the hybrid, the adapter card (in the following called hdmi module), the FEC and the virtual DAQ module which could be associated to the ethernet switch and the PC. It reflects the scalability of the system.

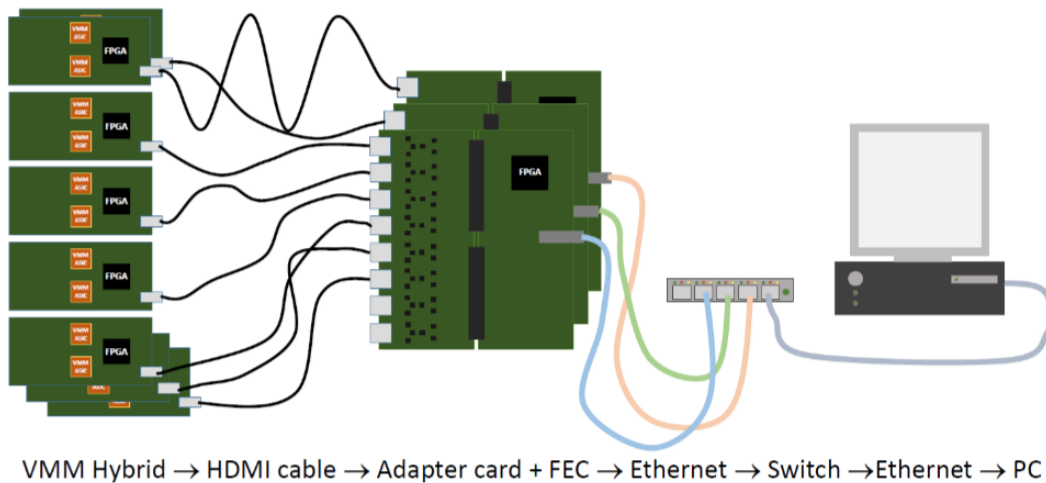


Figure 2: Schematic view of the detector electronics chain (components not to scale) [7]

2. Getting Started

The current release is the v 1.0 which can be cloned from the CERN gitlab repository. In order to install the software follow the instructions below.

2.1. Requirements

The software was tested on linux and requires the following packages:

- QT 5.7
- C++11 ($\text{gcc} \geq 4.7$)

2.2. Installation

2.2.1. Installing QT

If QT itself or its current version is not yet installed on your system, please follow the installation instruction on the [QT homepage](#).

The download and installation procedure should take less than an hour on a system with adequate internet connection.

2.2.2. Obtaining the software

As mentioned above the software can be checked out via the CERN gitlab repository. To do so type the following command in the terminal:

```
git clone https://:gitlab.cern.ch:8443/mguth/VMM-software-RD51.git
```

If the operation succeeded a new folder should appear with the name "VMM-software-RD51".

2.2.3. Compiling the software

Navigate to the following repository:

```
cd VMM-software-RD51/build
```

make sure you use the correct qmake in case you have several QT versions installed and run qmake:

```
qmake vmmdcs.pro
```

The next step is to compile the package via

```
make clean
```

```
make -j10
```

The executable will now be located in VMM-software-RD51/build.

To open the slow control use the following commands:

- macOS
./vmmdcs.app/Contents/MacOS/vmmdcs
- Linux
./vmmdcs

The vmmdcs.pro file is set up to point to the include paths / dependencies based on whether the operating system is linux or macOS (mac being the only other considered system other than linux).

3. Software Registers

The Graphical User Interface (GUI) of the current software version is shown in figure 3.

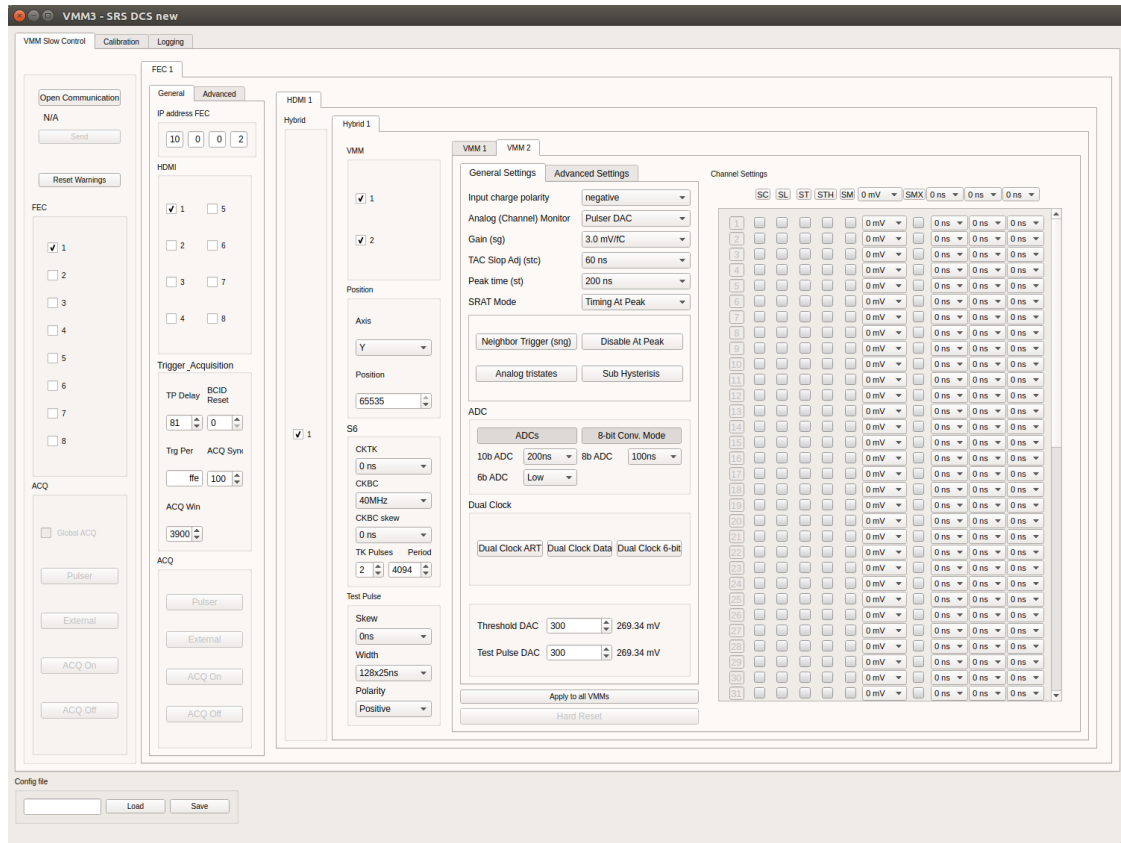
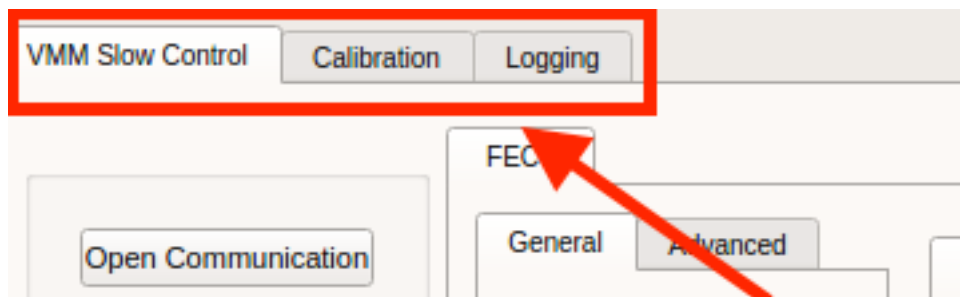


Figure 3: GUI of the VMM3 software v1.0

The structure of the new VMM3 software is divided into different levels. The three main registers are the

- VMM Slow Control,
- Calibration window,
- and the Logging window.



3.1. VMM Slow Control

The VMM slow control allows to modify the settings of the VMM chips, the Spartan 6 on the hybrid and the FEC.

The architecture of the software is designed such that the specific settings of each level are easily accessible. The resulting structure is shown in figure 4 and has the following registers:

1. DAQ level (red),
2. FEC level (blue),
3. HDMI level (green),
4. Hybrid level (purple),
5. VMM level (orange),
6. and the Config level (pink).

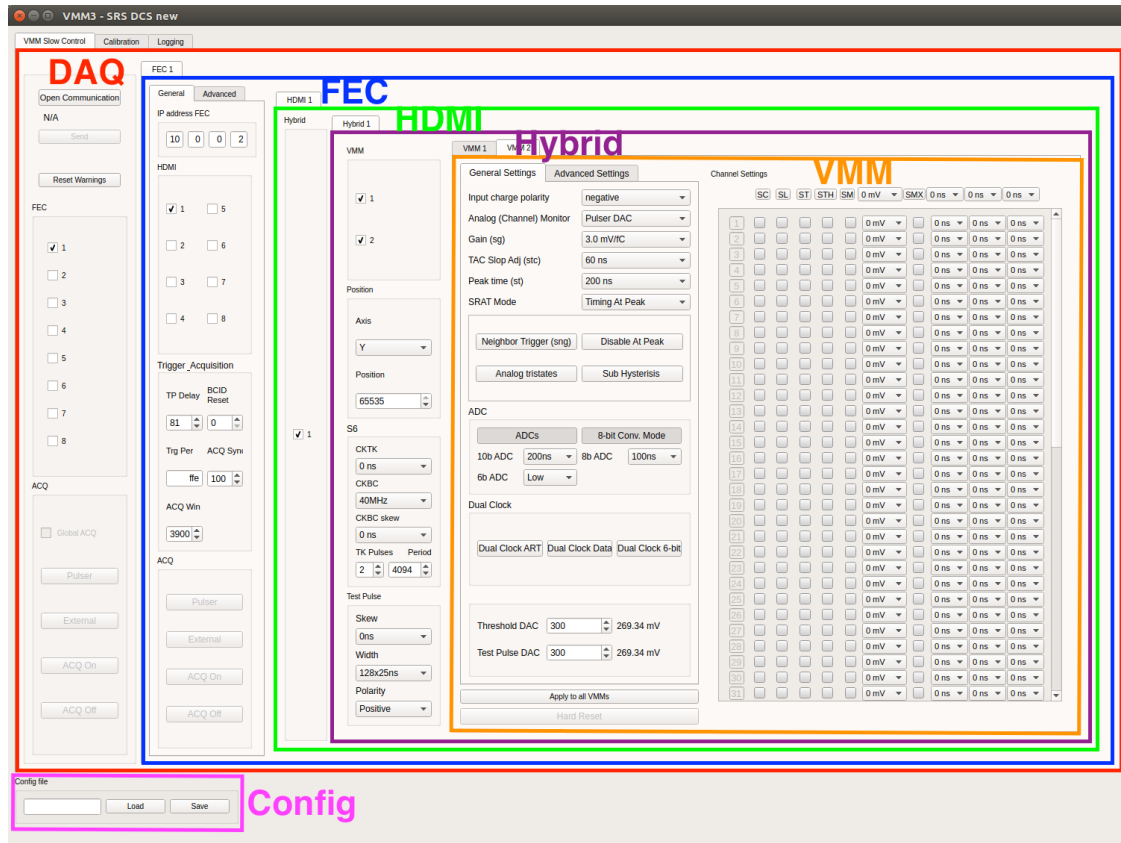


Figure 4: Registers of the GUI of the VMM3 software v1.0

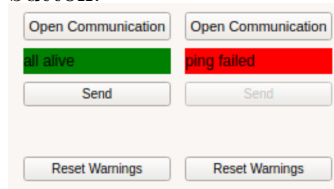
3.1.1. DAQ Level

The features of the DAQ level are shown in the figure on the right. There are no direct settings on this level since it does not directly correspond to a hardware level.

The purpose of this register is to check the connection to the activated FECs. In order to activate the FECs, check the boxes in the FEC field,



After setting the IP addresses of the single FECs as described in section 3.1.2 the communication can be verified via the **Open Communication** push button.

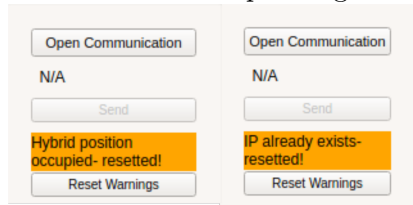


The response of the verification will be displayed under the push button as shown in the figure above. If the ping was successful for all activated FECs the text "all alive" will be shown but if the ping of at least one FEC fails "ping failed" will be displayed.

If "all alive" is displayed, further options in the lower levels will be enabled, more about this in the descriptions below, as well as the **Send** button.

The **Send** button sends out all settings of each sublevel to their corresponding IPs and ports. After changing any settings it is necessary to click the **Send** button to send the settings out. If this is not done no changes will be applied on the hardware level.

Between the **Send** and **Reset Warnings** button a warning display is located. Below two examples of warnings which will be explained in more detail in the corresponding sections.




The ACQ window on the DAQ level allows to enable the data acquisition of all FECs at once via a broadcast. If **Global ACQ** is checked all ACQ windows on FEC level are disabled. It is possible to chose between Pulser (continuous) and External trigger mode.

3.1.2. FEC Level

The settings on the FEC level are divided into two categories, **General** and **Advanced** as shown in the two figures on the right.

General Settings:

The IPv4 address of the FEC can be entered in the register **IP address FEC**. If the same IP address is entered for two FECs, a warning will be displayed on the DAQ level as described in the previous section.

To select the connected HDMI ports check the corresponding boxes in the HDMI field.

The Trigger Acquisition settings can be either modified by clicking on the up/down buttons or by entering the numbers directly whereas for the trigger period the number has to be entered in hexadecimals.

In the ACQ (acquisition) group box one can chose between the pulser and the external trigger mode. In addition, the data acquisition can be started via **ACQ On** and stopped via **ACQ Off**. These options are only available if the ping of all FECs was sucessfull and "all alive" is displayed on the DAQ level as described in section 3.1.1. By using the **ACQ On** button it is not necessary to send out the settings before also if some settings were changed since they are sent out automatically in this case.

Advanced Settings:

The L0 settings of the VMM can be changed on the FEC level for all connected VMMs at once.

The following features are also only available if the ping was sucessful:

- **Warnlimit FEC:** Softreset of FEC
- **Reboot FEC:** reboots the FEC
- **Reset Links:** Is disabled in the current version since it only crashes the whole FEC

The image displays two side-by-side screenshots of the 'FEC 1' configuration window, showing the 'General' and 'Advanced' tabs.

General Tab (Left):

- IP address FEC:** A field with the value '10.0.0.2'.
- HDMI:** A grid of checkboxes for ports 1 through 8. Port 8 is checked.
- Trigger Acquisition:**
 - TP Delay:** 81
 - BCID Reset:** 0
 - Trg Per:** ffff
 - ACQ Syn:** 100
 - ACQ Win:** 5000
- ACQ:** Buttons for 'Pulser', 'External', 'ACQ On', and 'ACQ Off'.

Advanced Tab (Right):

- L0 settings:**
 - sLOenaV:** (button)
 - sLOena:** (button)
 - IOffset:** 0
 - offset:** 0
 - rollover:** 0
 - window:** 0
 - truncate:** 0
 - nskip:** 0
 - sLOcktest:** (button)
- Frame Cnt:** (dropdown menu)
- HINFO+Dataler:** (dropdown menu)
- High Res.:** (checked checkbox)
- Buttons:** WarmInit FEC, Reboot FEC, Link Status, Reset Links.
- FEC Response:**
 - Clear:** (button)
 - Message:**

```
***** NEW PACKET
RECEIVED *****
Data received size:
24 bytes
Rec'd ID: 51
Data, 1: 8000
Data, 2: bbaaffff
Data, 3: 0
Data, 4: 0
Data, 5: 40004000
```

which has to be restarted.

- **Link Status:** Retrieves the status of the FEC as displayed in the right figure in the **FEC Response** box:

The most important information is the "Data 5" which provides informations about the connected hybrids to the FEC. The numbering is inverted, e.g. on the right figure above the hybrid 4 and 8 are connected with status 4.

- Status 2: Something connected but no connection
- Status 3: Connected but not ready to operate
- Status 4: Connected to hybrid, ready to operate
- Status 5: Connected and data acquisition is ongoing

3.1.3. HDMI Level

It is only possible to chose one hybrid at the moment on the HDMI level since it is not yet possible to connect two hybrids with each other as it is done with the APV25. This register is therefore meant for future updates of the software.

3.1.4. Hybrid Level

Again, on the hybrid level, the next lower levels can be chosen, in this case the VMMs. The layout is shown on the right.

In order to set the hybrid position on the detector, chose the axis and the corresponding position. The default position for the first chosen hybrid is 65535 on the Y axis which corresponds to the highest 16bit integer. For every additional hybrid the default value decreases by one.

The Spartan 6 settings can be changed in the **S6** group box.

The fourth category handles the Test Pulse on hybrid level.

The image shows a software configuration window for 'Hybrid 1'. It is divided into four main sections: VMM, Position, S6, and Test Pulse. The VMM section has two checkboxes, both of which are checked and labeled '1' and '2'. The Position section has a dropdown menu for 'Axis' set to 'Y' and a numeric input for 'Position' set to '65527'. The S6 section contains several settings: 'CKTK' is a dropdown set to '12.5 ns', 'CKBC' is a dropdown set to '160MHz', 'CKBC skew' is a dropdown set to '0 ns', and 'TK Pulses' and 'Period' are numeric inputs set to '2' and '4094' respectively. The Test Pulse section has three settings: 'Skew' is a dropdown set to '12.5ns', 'Width' is a dropdown set to '16x25ns', and 'Polarity' is a dropdown set to 'Negative'.

Section	Parameter	Value
VMM	1	☒
	2	☒
Position	Axis	Y
	Position	65527
S6	CKTK	12.5 ns
	CKBC	160MHz
	CKBC skew	0 ns
	TK Pulses	2
	Period	4094
Test Pulse	Skew	12.5ns
	Width	16x25ns
	Polarity	Negative

3.1.5. VMM Level

In the last level, the VMM settings can be accessed what allows to change the settings of the single VMM chips on every hybrid. The GUI of this level is shown in figure 5 and 6.

The screenshot displays the VMM Level GUI. At the top, there are tabs for 'VMM 1' and 'VMM 2'. Below these are three main sections: 'General Settings', 'Advanced Settings', and 'Channel Settings'.

General Settings: Includes dropdowns for 'Input charge polarity' (set to negative), 'Analog (Channel) Monitor' (set to Pulser DAC), 'Gain (sg)' (set to 3.0 mV/IC), 'TAC Slop Adj (stc)' (set to 60 ns), 'Peak time (st)' (set to 200 ns), and 'SRAT Mode' (set to Timing At Peak). There are also buttons for 'Neighbor Trigger (sng)', 'Disable At Peak', 'Analog tristates', and 'Sub Hysteresis'.

ADC: Features buttons for 'ADCs' and '8-bit Conv. Mode'. Below these are dropdowns for '10b ADC' (set to 200ns), '8b ADC' (set to 100ns), and '6b ADC' (set to Low).

Dual Clock: Includes buttons for 'Dual Clock ART', 'Dual Clock Data', and 'Dual Clock 6-bit'.

Threshold DAC: A slider set to 300, corresponding to 269.34 mV.

Test Pulse DAC: A slider set to 300, corresponding to 269.34 mV.

Channel Settings: A table with 31 rows (numbered 1 to 31) and 10 columns. The columns are labeled SC, SL, ST, STH, SM, and then three columns of voltage and time settings (0 mV, 0 ns, 0 ns). Each cell in the table contains a dropdown menu, all of which are currently set to 0 mV or 0 ns.

At the bottom of the GUI, there are two buttons: 'Apply to all VMMs' and 'Hard Reset'.

Figure 5: GUI of software v1.0 of VMM level, on the left general settings and on the right the channel settings for channel 1 to 31.

In figure 5 the general settings of the VMM are shown and in figure 6 the advanced settings, also the channels differ in the two figures.

The corresponding settings of the VMM3 are shown in the figures 9, 10, 11 and 12 extracted from the VMM3 manual [5].

The button **Apply to all VMMs** on the bottom can be used to apply all settings (except the single channel specific ones) to all other activated VMMs.

The **Hard Reset** of the VMM is only available if **Open Communication** on the DAQ level was successful (for more details see 3.1.1). In order to be sure that the VMMs receive the settings

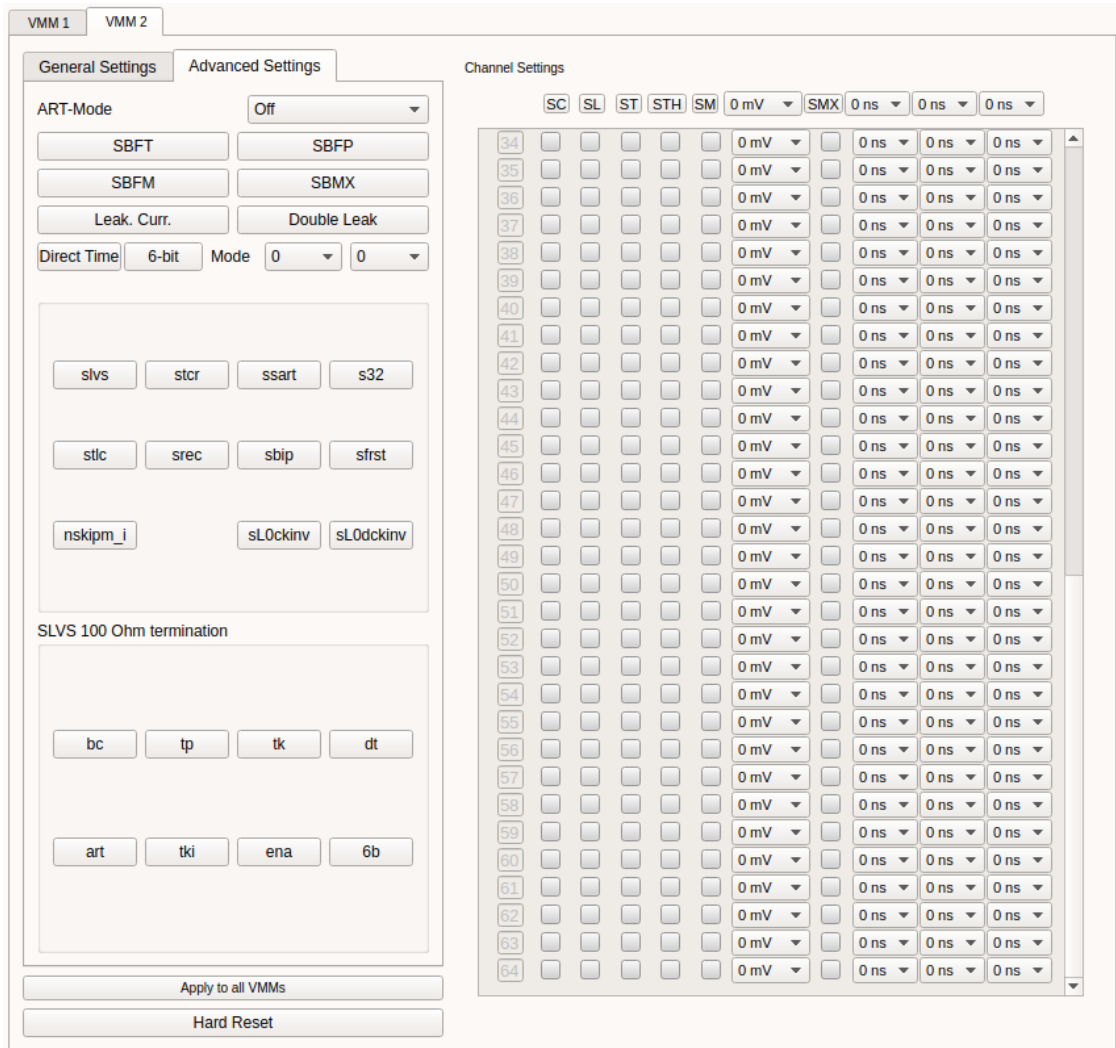


Figure 6: GUI of software v1.0 of VMM level, on the left advanced settings and on the right the channel settings for channel 34 to 64.

and also the reset instruction, use the button **Link Status** on the FEC as described in section 3.1.2.

One special feature is the **Analog (Channel) Monitor** which allows the analog monitoring of the Pulser DAC etc. as well as the monitoring of single channels. The switch of the *scmx* is done automatically. If the config files (see section 3.1.6) are directly used to change the settings instead of the GUI use for the monitoring settings the following options:

- Pulser DAC: 64,
- Threshold DAC: 65,
- Bandgap Reference: 66,
- Temperature Sensor: 67,

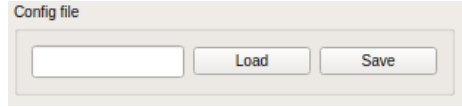
- and for the Channels: 0-63, corresponding to the channels 1-64.

In the config file the convention for the channels is that they start with 0 but in the GUI they start with 1, be careful by comparing them!

To see the analog output on the VMM pins it is necessary to enable the **SBFM** setting which can be found under advanced settings.

3.1.6. Config Module

The config module as shown in the following figure handles the configuration files.



These config files can be used to store the settings made in the GUI. To do so, chose a name of the files (do not enter file extension) and press **Save**. This will generate one config file which contains the structure i.e. which FECs, HDMIs, etc. are active and for each active register an additional config file will be created. To load the settings type the name of your config file (only general name, not the one of the single registers) and press **load**.

It is possible to load a default file when starting the Slow Control. This is possible by creating the config file *default*. Set up all your settings in the GUI and save the config file with the name *default* as described above.

3.2. Calibration

For the moment only two calibrations are included in the software, the ADC and the TDC calibration. Later, more calibrations will follow for the threshold etc. but another ADC module on the hybrid is required for this purpose.

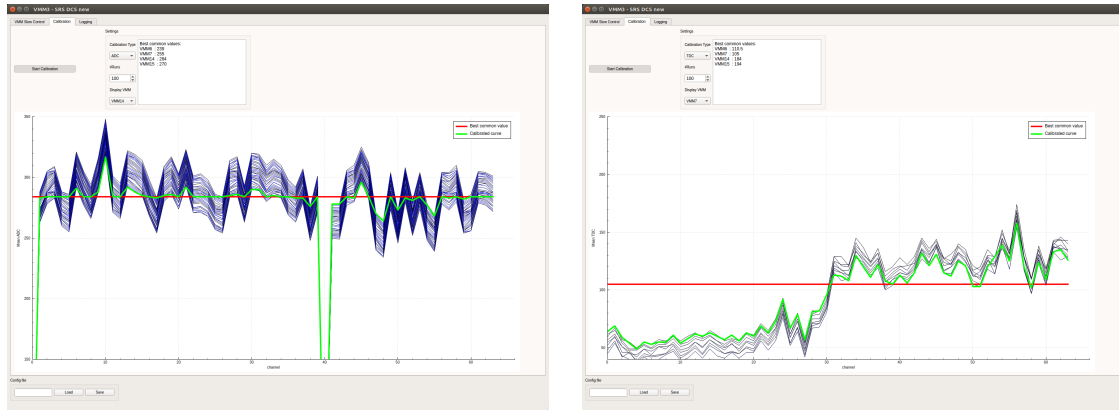


Figure 7: The left figure shows the ADC calibration, with green the calibrated curve, and the on the right the TDC calibration.

The current version of the calibration is not yet finalised. As shown in figure 7 the calibration is not able to achieve one common value for all channels. At the moment the largest overlap between all channels is taken into account to calibrate to a common value. The value for the

outlying channels are set to the closest possible value to the calibration value.

The ADC calibration gives better results than the TDC calibration. At the end, one has to decide to either improve the algorithm or to simply store calibration factors for each channel separately in order to take them into account in the offline analysis.

3.3. Logging Window

In this window, several informations are displayed which can help to identify problems or just to know what the code is doing. Enabling the **Debug** button allows an even more detailed (debug) output.

4. Examples

For simplicity, please use the following setup to use the examples:

- FEC 1
- HDMI 1
- Hybrid 1
- VMM 1 and 2

4.1. Test Pulse Setup

The VMM provides a possibility to generate test pulses and read them out. An example setup to create these test pulses is in the example configs. These settings can be accessed in the config module.

Type into the text box `examples/TestPulse` and click on **Load**. This is only an example setting and does not necessarily correspond to your setup. First, correct the IP address of the FEC if necessary. Then, check if the activated HDMI (in the example case 1) corresponds to your setup, if not check also your HDMI.

Afterwards click on the button **Open Communication** on the DAQ level. If the ping succeeded and "all alive" is displayed chose the pulser mode on the FEC level. The test pulse can be started by clicking on **ACQ On**. The test pulse can be displayed in an oscilloscope as shown in figure 8.

In case that the config file is not available a summary of the settings:

- Enable ST setting of single channels which should have a test pulse,
- enable SBFM for analog monitoring to display test pulses on oscilloscope,
- optional:
 - Varying Spartan 6 (S6) settings on hybrid level,
 - changing Trigger_Acquisition settings on FEC level

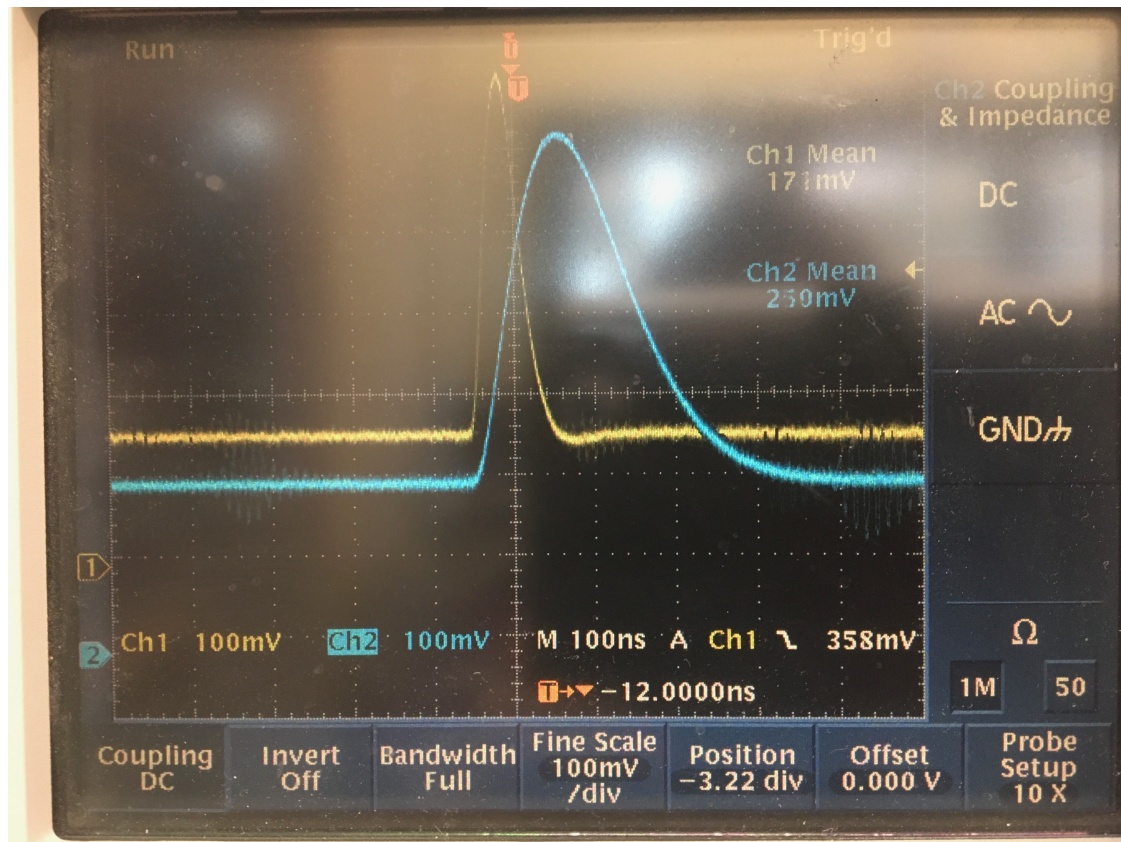


Figure 8: Test pulses on the oscilloscope

References

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- [2] ATLAS MUON Collaboration. New small wheel technical design report. *Tech. Des. Rep. CERN/LHCC-2013-006*, CERN, 2013.
- [3] ATLAS Collaboration. ATLAS detector and physics performance Technical Design Report, Volume II. 1999.
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- [6] Hans Muller. *SRS Scalable Readout System*. https://indico.cern.ch/event/240381/contributions/1554636/attachments/406595/564871/SRS_readout_systems.pdf.
- [7] Michael Lupberger. Brightness - building a research infrastructure and synergies for highest scientific impact on ess. Technical report, D4.9 Detector electronics chain.

A. VMM Resgister details

Register Type	Sequence
global registers - first bank	sp sdp sbmx sbft sbfp sbfm slg sm5:sm0 scmx sfa sfam st1:0 sfm sg2:0 sng stot sttt ssh stc1:0 sdt9:0 sdp9:0 sc010b:sc110b sc08b:sc18b sc06b:sc26b s8b s6b s10b sdcks sdcka sdck6b sdrv stpp res00 res0 res1 res2 res3 slvs s32 stcr ssart srec stlc sbip srat sfrst slvsbc slvstp slvstk slvsdt slvsart slvstki slvsena slvs6b sL0enaV nu nu nu nu nu nu nu nu reset reset
channel register (64×)	sc sl st sth sm smx sd0:sd4 sz010b:sz410b sz08b:sz38b sz06b:sz26b nu
global registers - second bank	nu0:30 nskipm_i sL0cktest sL0dckinv sL0ckinv sL0ena truncate_i0:5 nskip_i0:6 window_i0:2 rollover_i0:11 l0offset_i0:11 offset_i0:11 nu0:7

Figure 9: Sequence of Configuration Registers of VMM3.[5]

Channel bits (defaults are 0)	Description
sc [0 1]	large sensor capacitance mode ([0] <~200 pF , [1] >~200 pF)
sl [0 1]	leakage current disable [0=enabled]
st [0 1]	300 fF test capacitor [1=enabled]
sth [0 1]	multiplies test capacitor by 10
sm [0 1]	mask enable [1=enabled]
sd0-sd4 [0:0 through 1:1]	trim threshold DAC, 1 mV step ([0:0] trim 0 V , [1:1] trim -29 mV)
smx [0 1]	channel monitor mode ([0] analog output, [1] trimmed threshold))
sz010b, sz110b, sz210b, sz310b, sz410b	10-bit ADC zero
sz08b, sz18b, sz28b, sz38b	8-bit ADC zero
sz06b, sz16b, sz26b	6-bit ADC zero

Figure 10: Channel Configuration Registers of VMM3.[5]

Global bits (defaults are 0)	Description
sp	input charge polarity ([0] negative, [1] positive)
sdp	disable-at-peak
sbmx	routes analog monitor to PDO output
sbft [0 1], sbfp [0 1], sbfm [0 1]	analog output buffers, [1] enable (TDO, PDO, MO)
slg	leakage current disable ([0] enabled)
sm5-sm0, scmx	monitor multiplexing. - Common monitor: scmx, sm5-sm0 [0 000001 to 000100], pulser DAC (after pulser switch), threshold DAC, band-gap reference, temperature sensor) - channel monitor: scmx, sm5-sm0 [1 000000 to 111111], channels 0 to 63
sfa [0 1], sfam [0 1]	ART enable (sfa [1]) and mode (sfam [0] timing at threshold, [1] timing at peak)
st1,st0 [00 01 10 11]	peaktime (200, 100, 50, 25 ns)
sfm [0 1]	enables dynamic discharge for AC coupling ([1] enable)
sg2,sg1,sg0 [000:111]	gain (0.5, 1, 3, 4.5, 6, 9, 12, 16 mV/fC)
sng	neighbor (channel and chip) triggering enable
stot [0 1]	timing outputs control 1 (s6b must be disabled) - stpp,stot[00,01,10,11]: TtP,ToT,PtP,PtT - TtP: threshold-to-peak - ToT: time-over-threshold - PtP: pulse-at-peak (10ns) (not available with s10b) - PtT: peak-to-threshold (not available with s10b)
sttt [0 1]	enables direct-output logic (both timing and s6b)
ssh [0 1]	enables sub-hysteresis discrimination
stc1,stc0 [00 01 10 11]	TAC slope adjustment (60, 100, 350, 650 ns)
sdt9-sdt0 [0:0 through 1:1]	coarse threshold DAC
sdp9-sdp0 [0:0 through 1:1]	test pulse DAC
sc010b,sc110b	10-bit ADC conversion time
sc08b,sc18b	8-bit ADC conversion time
sc06b, sc16b, sc26b	6-bit ADC conversion time
s8b	8-bit ADC conversion mode
s6b	enables 6-bit ADC (requires sttt enabled)
s10b	enables high resolution ADCs (10/8-bit ADC enable)
sdcks	dual clock edge serialized data enable
sdcka	dual clock edge serialized ART enable
sdck6b	dual clock edge serialized 6-bit enable
sdrv	tristates analog outputs with token, used in analog mode
stpp [0 1]	timing outputs control 2
slvs	enables direct output IOs
stcr	enables auto-reset (at the end of the ramp, if no stop occurs)
ssart	enables ART flag synchronization (trail to next trail)

Figure 11: Global Configuration Registers 1 of the VMM3.[5]

Global bits (defaults are 0)	Description
s32	skips channels 16-47 and makes 15 and 48 neighbors
stlc	enables mild tail cancellation (when enabled, overrides sbip)
srec	enables fast recovery from high charge
sbip	enables bipolar shape
srat	enables timing ramp at threshold
sfrst	enables fast reset at 6-b completion
slvsbc	enable slvs 100 Ω termination on ckbc
slvstp	enable slvs 100 Ω termination on cktp
slvstk	enable slvs 100 Ω termination on ctkk
slvsdt	enable slvs 100 Ω termination on ckdt
slvsart	enable slvs 100 Ω termination on ckart
slvstki	enable slvs 100 Ω termination on ctkki
slvsena	enable slvs 100 Ω termination on cken
slvs6b	enable slvs 100 Ω termination on ck6b
sL0enaV	disable mixed signal functions when L0 enabled
reset reset	Hard reset when both high
sL0ena	enable L0 core / reset core & gate clk if 0
l0offset_i0:11	L0 BC offset
offset_i0:11	Channel tagging BC offset
rollover_i0:11	Channel tagging BC rollover
window_i0:2	Size of trigger window
truncate_i0:5	Max hits per L0
nskip_i0:6	Number of L0 triggers to skip on overflow
sL0cktest	enable clocks when L0 core disabled (test)
sL0ckinv	invert BCCLK
sL0dckinv	invert DCK
nskipm_i	magic number on BCID - 0xFE8

Figure 12: Global Configuration Registers 2 of the VMM3.[5]